
Attention Circuits, Not Weights: What Actually Transfers in Pretrained Vision Transformers

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Abstract

Understanding what fine-tuning actually inherits from a pretrained Vision Transformer remains an open question. We address it through a systematic elimination study spanning over twenty ablation conditions, progressively stripping components from pretrained weights to locate the source of the transfer benefit. The elimination converges on a precise answer: the primary carrier of transfer is the *attention circuit*—the pairwise product of query–key matrices (controlling where to attend) and the pairwise product of value–projection matrices (controlling what to extract) within each late-layer head. MLP weights, early-layer weights, single-matrix spectral structure, and even the element-wise pretrained values themselves are all dispensable. Based on this finding, we propose *circuit transfer*: we extract these two products from selected heads, refactor each into a fresh pair of weight matrices via SVD, and use the result as initialisation. The refactored weights preserve the head’s computation exactly while discarding source-task scaffolding. Applied to the last four transformer blocks, circuit transfer matches or exceeds full pretraining on Tiny-ImageNet (60.87% vs. 60.65%) and CIFAR-100 (75.52% vs. 75.03%); at ImageNet-1K scale with ViT-Base, it recovers 47% of the full pretraining gain. Decomposing the two circuits further reveals a striking asymmetry: the value–projection side accounts for roughly four times the benefit of the query–key side, inverting the field’s QK-centric focus. Trajectory analysis during pretraining uncovers a two-phase formation process—rapid rank collapse followed by abrupt query–key alignment—that completes within the first 20% of training. Finally, applying the same circuit-level diagnostics to MAE, DINO, and DINOv2 shows that all three methods install the same structural signatures, establishing attention circuits as a general account of what pretraining transfers rather than an artefact of our controlled probe. Code is anonymously available at this [link](#).

1 Introduction

Pretraining followed by fine-tuning has become the default recipe for Vision Transformers [Dosovitskiy et al., 2021, Touvron et al., 2021], mirroring its earlier consolidation in convolutional networks [He et al., 2016, Yosinski et al., 2014] and underpinning the broader paradigm of foundation models [Bommasani et al., 2021]. Across self-supervised methods as different as masked reconstruction [He et al., 2022], self-distillation [Caron et al., 2021, Oquab et al., 2024], and procedural tasks without any visual content [Shinnick et al., 2025], pretraining consistently improves downstream accuracy over training from scratch. This robustness to the choice of pretext task is striking, but it also leaves a fundamental question unanswered: *what, mechanistically, does a pretrained ViT pass to its fine-tuned descendant?*

Existing analyses identify properties that correlate with the transfer benefit—improved spectral structure of weights [Arora et al., 2019, Martin and Mahoney, 2021], flatter loss land-

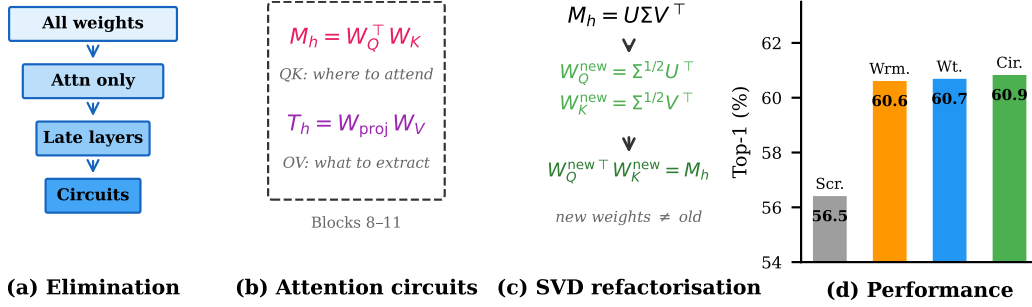


Figure 1: **Overview.** (a) Systematic elimination progressively strips components from pretrained weights to locate the transfer benefit. (b) The elimination converges on the QK and OV product matrices of late-layer attention heads. (c) These products are refactored via SVD into fresh weight matrices that preserve each head’s computation while discarding source-task scaffolding. (d) Top-1 accuracy: Scr. = scratch (Xavier init), Wrm. = full warmup, Wt. = attention weight transfer (L8–11), Cir. = circuit transfer (ours). Circuit transfer matches or exceeds full pretraining despite discarding all MLP weights, all early-layer weights, and all element-wise weight values.

scapes [Neyshabur et al., 2020], higher effective rank [Kumar et al., 2022], more structured attention patterns [Raghu et al., 2021]—but each captures only a partial aspect of the picture, and none yields an actionable account of which pretrained components are essential, which are redundant, and which can actively hurt. Our goal is to provide such an account by identifying the *primary carrier* of pretrained information: the component whose presence is necessary for the transfer benefit and whose isolation still produces a substantial gain.

We approach this through an *exhaustive elimination*: starting from a fully pretrained ViT, we progressively strip away components—spectral structure, feature directions, individual modules, layer ranges—and measure when the transfer benefit disappears. The elimination converges on a precise answer: the transferable signal lives in the *attention circuit* of each late-layer head. Following Elhage et al. [2021], each head’s computation is governed by two product matrices,

$$\begin{aligned} M_h &= W_{Q_h}^\top W_{K_h} && \text{(the QK circuit, what to attend to),} \\ T_h &= W_{\text{proj}_h} W_{V_h} && \text{(the OV circuit, what to extract),} \end{aligned}$$

which together fully determine the head’s input-output behaviour. Crucially, any factorisation of the same products produces the same function. MLP weights, early-layer weights, single-matrix spectral structure, and even the element-wise pretrained values themselves are all dispensable—the circuit products alone carry the transferable information.

This algebraic invariance directly suggests a transfer mechanism. We introduce *circuit transfer*: extract M_h and T_h from selected heads of the pretrained model, refactor each into a fresh pair of weight matrices via SVD (Eq. 2), and use the result as initialisation. The refactored weights differ element-wise from the originals but realise the same circuits, while resetting source-task scaffolding to a Xavier-compatible form. Applied to just blocks 8–11, circuit transfer matches or exceeds full pretraining on Tiny-ImageNet (60.87% vs. 60.65%) and CIFAR-100 (75.52% vs. 75.03%). At ImageNet-1K scale with ViT-Base, it recovers 47% of the full pretraining gain (+0.81% over scratch, against +1.78% for full warmup).

The same framework answers two further questions. By transferring each circuit in isolation, we find that the OV circuit dominates the QK circuit by roughly four-to-one—a striking inversion of the field’s QK-centric focus in attention analysis. By tracking circuit metrics across pretraining checkpoints, we observe a two-phase formation process: a rapid rank collapse in the first 1,000 steps, followed by an abrupt W_Q - W_K alignment between steps 2,500 and 3,000, after which the structure is essentially fixed for the remaining 80% of training. Finally, applying the same diagnostics to MAE, DINO, and DINOv2 reveals that all three exhibit the same circuit-level signatures, confirming that attention circuits describe what mainstream pretraining methods install—not just our controlled probe.

In summary, we make four contributions: (i) we identify attention circuits in late-layer heads as the primary carrier of pretraining transfer, through systematic elimination across 20+ ablation conditions

(Section 3); (ii) we introduce circuit transfer via SVD refactorisation and show it matches or exceeds full pretraining on medium-complexity benchmarks while recovering 47% of the gain at ImageNet-1K scale (Section 3); (iii) we show that OV circuits dominate QK circuits by four-to-one, and that circuit structure emerges in two distinct phases within the first 20% of pretraining (Section 4); and (iv) we demonstrate that MAE, DINO, and DINOv2 share the same circuit-level signatures, establishing the framework as a general account of pretraining transfer (Section 4).

2 Related Work

A long line of work characterises pretrained weights through their spectral properties: Arora et al. [2019] link implicit regularisation to the singular structure of deep matrix factorisations, Martin and Mahoney [2021] relate heavy-tailed singular value distributions to model quality, and Kumar et al. [2022] show that fine-tuning can distort pretrained features. These analyses predict that the transferable benefit should be recoverable from each weight matrix’s spectrum alone, but our elimination shows that this prediction fails: preserving the spectrum while randomising directions captures at most a quarter of the gain (Section 3.3). The transferable structure is not located in any single matrix’s SVD, but in the *products* of matrices. SVD-based parameterisations of attention have also been used for compression [Hsu et al., 2022] and parameter-efficient adaptation [Hu et al., 2022], and sparsification studies such as the Lottery Ticket Hypothesis [Frankle and Carbin, 2019] similarly seek small subnetworks that capture the trainable benefit; our results (Section 4.1) suggest the relevant transfer structure is not low-rank, in tension with these compression-based intuitions.

The framework we build on, due to Elhage et al. [2021], treats a transformer head as a pair of product matrices—the QK circuit $W_Q^\top W_K$ and the OV circuit $W_{\text{proj}} W_V$ —that fully determine the head’s input-output behaviour. This decomposition has been used as an analytical lens for language model behaviour, including the discovery of induction heads [Olsson et al., 2022] and head specialisation studies [Clark et al., 2019, Voita et al., 2019], and it complements geometric analyses of attention such as the relation between attention and convolution [Cordonnier et al., 2020]. These studies have generally focused on the QK side of the circuit (attention patterns, positional biases). We bring the framework to vision transformers, repurpose it from analysis to a transfer mechanism, and find that the OV side—usually the quieter half of the analysis—is in fact the dominant carrier of transferable information. Complementary work studies what transfers behaviourally, attributing benefit to flatness of the loss landscape [Neyshabur et al., 2020, Foret et al., 2021], comparing representations through CKA [Kornblith et al., 2019, Raghu et al., 2021], or analysing the inductive biases of ViT representations [Naseer et al., 2021, Park and Kim, 2022, Geirhos et al., 2019]. These characterise the *state* of the pretrained model but do not isolate the minimal component responsible for the benefit.

Vision transformer pretraining itself has matured into a rich design space, with masked-image methods adapted from masked language modelling [Devlin et al., 2019]—MAE [He et al., 2022] and BEiT [Bao et al., 2022]—self-distillation [Caron et al., 2021, Oquab et al., 2024], contrastive training [Chen et al., 2021], and architectural variants such as Swin [Liu et al., 2021]. Our circuit-level analysis applies uniformly across these methods (Section 4.2). Our pretraining probe follows Shinnick et al. [2025], who show that procedural pretraining on context-free grammars (such as the DYCK languages) produces measurable downstream gains on real-image classification despite the absence of any visual content. We use this setting because it makes the source and downstream tasks fully disjoint at the feature level, and we replicate their ViT-Base ImageNet-1K baselines within 0.3% before introducing circuit transfer on top.

3 Attention Circuits and the Transfer Unit

We begin by introducing the algebraic framework that makes our analysis precise, then use it to locate the transferable signal through systematic elimination.

3.1 Circuit framework

A self-attention head is conventionally described by four weight matrices, $W_Q, W_K, W_V, W_{\text{proj}}$, but its computational behaviour depends only on two products of these matrices. This algebraic redundancy is the entry point of our analysis: it allows us to ask whether what transfers in pretraining

is the weight values themselves or the products they realise. For each head h in a transformer block, the standard attention computation [Vaswani et al., 2017, Elhage et al., 2021] on input $X \in \mathbb{R}^{n \times d}$ is

$$\text{Attn}_h(X) = \text{softmax}\left(\frac{X W_{Q_h}^\top W_{K_h} X^\top}{\sqrt{d_k}}\right) X W_{V_h} W_{\text{proj}_h}^\top, \quad (1)$$

which is governed by two product matrices with distinct functional roles: the QK circuit $M_h = W_{Q_h}^\top W_{K_h} \in \mathbb{R}^{d \times d}$, controlling *where to attend*, and the OV circuit $T_h = W_{\text{proj}_h} W_{V_h} \in \mathbb{R}^{d \times d}$, controlling *what to extract* (the output of Eq. 1 is $\text{softmax}(\dots) X W_{V_h} W_{\text{proj}_h}^\top$, so the effective value-space map is $T_h^\top = W_{V_h} W_{\text{proj}_h}^\top$; we follow the convention of Elhage et al. [2021] and define $T_h = W_{\text{proj}_h} W_{V_h}$ for notational symmetry with M_h). Together they fully determine the head’s input-output behaviour, and any pair (W_{Q_h}', W_{K_h}') with $W_{Q_h}'^\top W_{K_h}' = M_h$ produces the same attention pattern, regardless of element-wise differences in the underlying weights.

This algebraic invariance suggests a direct way to test what transfers in pretraining: extract M_h and T_h from a pretrained model, then refactor each into a fresh pair of weight matrices via singular value decomposition (SVD). Writing $M_h = U_M \Sigma_M V_M^\top$ and $T_h = U_T \Sigma_T V_T^\top$, we set

$$W_{Q_h}^{\text{new}} = \Sigma_M^{1/2} U_M^\top, \quad W_{K_h}^{\text{new}} = \Sigma_M^{1/2} V_M^\top, \quad W_{V_h}^{\text{new}} = \Sigma_T^{1/2} V_T^\top, \quad W_{\text{proj}_h}^{\text{new}} = U_T \Sigma_T^{1/2}. \quad (2)$$

By construction, $W_{Q_h}^{\text{new}^\top} W_{K_h}^{\text{new}} = M_h$ and $W_{\text{proj}_h}^{\text{new}} W_{V_h}^{\text{new}} = T_h$ exactly: the refactored weights differ element-wise from the original pretrained weights—in scale, in spatial pattern, and in individual values—but the head computes precisely the same function. The symmetric splitting through $\Sigma^{1/2}$ also yields singular value distributions compatible with Xavier-style initialisation, which proves important for stable downstream training [Glorot and Bengio, 2010]. We call a model initialised in this way—using Eq. 2 for selected heads, and Xavier initialisation elsewhere—a *circuit-transfer* model.

3.2 Experimental setup

The circuit framework provides a precise way to separate a head’s functional computation from its specific weight parameterisation, but it does not tell us *which* circuits carry the transferable signal. We identify this empirically through systematic elimination.

We use a ViT-Tiny (12 layers, 3 heads, 192-dim embedding, patch size 16) pretrained on the DYCK-64 balanced-parenthesis grammar [Shinnick et al., 2025] via masked token prediction for 15,000 steps (batch size 256, cosine LR schedule with peak 5×10^{-4}). This procedural pretraining is deliberately chosen for its complete absence of visual content: any downstream benefit on images must arise from structural properties of the learned weights rather than from feature-level overlap. All image classification experiments follow the standard DeiT recipe [Touvron et al., 2021, Steiner et al., 2022]: AdamW [Loshchilov and Hutter, 2019] (lr = 2×10^{-3} , weight decay 0.05), cosine schedule with 50-epoch warmup, Mixup [Zhang et al., 2018] ($\alpha=0.8$), CutMix [Yun et al., 2019] ($\alpha=1.0$), label smoothing (0.1) and RandAugment [Cubuk et al., 2020], for 300 epochs with batch size 512 on 8 GPUs (Distributed-DataParallel). Models are implemented in `timm` [Wightman, 2019]. We report top-1 validation accuracy on Tiny-ImageNet-200 [Le and Yang, 2015] and CIFAR-100 [Krizhevsky, 2009] for the elimination study, and on ImageNet-1K [Deng et al., 2009] with ViT-Base for scale validation. Table 1 lists the full hyperparameters.

Table 1: **Hyperparameters** (DeiT recipe).

Parameter	Value
Optimiser	AdamW
Base learning rate	2×10^{-3}
Weight decay	0.05
LR schedule	Cosine to 10^{-5}
Warmup epochs	50 (linear)
Total epochs	300
Batch size	512 (8 GPUs)
Gradient clipping	1.0
Mixup / CutMix	0.8 / 1.0
Label smoothing	0.1
RandAugment	2 ops, mag. 9
Image size	224×224

3.3 Elimination and circuit transfer

We initialise the model from the pretrained weights but progressively strip components, asking at each step whether the remaining structure still benefits downstream performance. Table 2 summarises the full set of 20+ ablation conditions.

Table 2: **Complete elimination results** on Tiny-ImageNet (ViT-Tiny, 300 epochs, 8-GPU DDP). “% Gain” is the fraction of the full-pretraining improvement (+4.20%) recovered by each condition.

Condition	Description	Top-1 (%)	Δ	% Gain
scratch	Xavier baseline	56.45	—	0
warmup	Full DYCK warmup	60.65	+4.20	100
<i>Single-matrix spectral statistics:</i>				
sbi_full	Warmup σ , random U, V	57.4	+0.95	23
sbi_uv	Warmup U, V , Xavier σ	56.3	−0.15	0
sbi_norm	Matched Frobenius norm	57.1	+0.65	15
sbi_rank	Matched effective rank	56.1	−0.35	0
<i>Module-level selection:</i>				
attn_only	All attn weights, all layers	58.1	+1.65	39
mlp_only	All MLP weights	56.1	−0.35	0
<i>Layer-level selection:</i>				
warmup_early	All weights L0–3	55.3	−1.15	−27
warmup_mid	All weights L4–7	56.8	+0.35	8
warmup_late	All weights L8–11	59.4	+2.95	70
<i>Precision targeting (attention circuits):</i>				
attn_late	Attn weights L8–11	60.73	+4.28	102
attn_late_qkv	QKV only L8–11	59.1	+2.65	63
qk_product_late	QK circuit L8–11	57.15	+0.70	17
ov_product_late	OV circuit L8–11	59.44	+2.99	71
attn_product_late	QK+OV circuit L8–11	60.87	+4.42	105
attn_product_all	QK+OV circuit all layers	58.75	+2.30	55
<i>Synthetic controls:</i>				
lowrank_qk	Synthetic low-rank QK (erank=33)	56.4	−0.05	0

Four observations follow. First, spectral information is far from sufficient: preserving each weight matrix’s singular spectrum captures at most a quarter of the gain (sbi_full), and preserving directions while randomising magnitudes captures none (sbi_uv). The transferable signal does not live in any single matrix’s SVD. Second, MLP weights are equally inert (mlp_only). Third, early layers are not merely useless but *harmful* (warmup_early): blocks 0–3 carry DYCK-specific token processing that actively conflicts with image patches. Fourth, late-layer attention recovers essentially the entire pretraining benefit (attn_late: 102%). Quantitatively, the gradient from spectral controls (23%) to attention-only (39%) to late-layer attention (102%) reveals that the signal grows as we narrow the component set. Notably, warmup_late (70%) includes MLP weights alongside attention, yet adding them *reduces* the gain from 102%—MLP weights from the pretrained model introduce scaffolding that conflicts with image fine-tuning. The synthetic control lowrank_qk (0%) further confirms that merely matching the pretrained rank structure yields no benefit; the specific learned subspace is what matters.

The transfer is thus confined to the attention mechanism of the last four blocks—but the elimination alone cannot resolve *what about these weights* matters, since single-matrix spectral properties have already failed.

We answer this with the circuit refactorisation of Section 3.1. We extract M_h and T_h from every head in blocks 8–11 of the pretrained model, refactor each into fresh weights via Eq. 2, and use the result as initialisation for image classification. All weights outside blocks 8–11—MLP weights, early-layer attention, embeddings, classification head—are left at Xavier initialisation; no element-wise pretrained values are copied.

Table 3 reports the result. Circuit transfer exceeds full pretraining by +0.22% on Tiny-ImageNet and by +0.49% on CIFAR-100, and exceeds raw weight transfer in both cases. The intuition is that pretrained weights carry both the functional computation we want (captured by M_h and T_h) and source-task scaffolding we do not (weight magnitudes tuned for DYCK tokens, momentum traces from the pretraining optimiser, LayerNorm-adapted scales). Direct copying preserves both; circuit transfer preserves only the former and—through the $\Sigma^{1/2}$ symmetric splitting—resets the latter to a Xavier-compatible form. This is the headline result: attention circuits are the *primary carrier* of

Table 3: **Circuit transfer vs. weight transfer** on two medium-complexity benchmarks (ViT-Tiny, 300 epochs). Circuit transfer matches or exceeds full pretraining despite discarding all MLP weights, all early-layer weights, and all element-wise weight values.

Method	Tiny-ImageNet (%)	CIFAR-100 (%)
Scratch (Xavier init)	56.45	72.82
Full warmup transfer	60.65	75.03
Weight transfer (attention weights, L8–11)	60.73	75.08
Circuit transfer (QK+OV, L8–11)	60.87	75.52

Table 4: **CIFAR-100 cross-validation** (ViT-Tiny, 300 epochs, 8-GPU DDP). Circuit transfer recovers 122% of the warmup gain, an even larger margin than on Tiny-ImageNet (105%).

Method	Top-1 (%)	Δ Scratch	% Gain
Scratch	72.82	—	0
Full warmup	75.03	+2.21	100
Weight transfer (attn, L8–11)	75.08	+2.26	102
Circuit transfer (QK+OV, L8–11)	75.52	+2.70	122

pretraining transfer—the component that accounts for the bulk of the benefit on medium-complexity benchmarks, and from which raw weights deviate only in ways that hurt.

Table 4 cross-validates this finding on CIFAR-100, where the circuit transfer advantage is even more pronounced: 122% of the full warmup gain, compared with 105% on Tiny-ImageNet. The larger margin on the simpler dataset is consistent with the intuition above: the less complex the downstream task, the more harmful the source-task scaffolding becomes, and the more circuit transfer benefits from discarding it.

Training curves (Figure 2) confirm that circuit transfer maintains a consistent advantage over scratch throughout training, ruling out early-stopping as an explanation.

3.4 Scale validation: ImageNet-1K with ViT-Base

To test whether these findings survive at production scale, we repeat the comparison with ViT-Base on ImageNet-1K. The DYCK warm-up phase follows the recipe of Shinnick et al. [2025], allowing us to validate our pipeline by comparing scratch and full-warmup baselines directly with their published numbers.

Two findings emerge from Table 5. First, circuit transfer recovers nearly half of the full pretraining gain (+0.81% out of +1.78%, or 47%) using only the QK and OV product matrices of four attention layers, while plain attention copying recovers only 26%—showing that SVD refactorisation conveys structural information that raw copying does not. Second, full warmup remains best in absolute accuracy. This is consistent with the circuit interpretation rather than at odds with it: on medium-complexity benchmarks, source-task scaffolding outweighs whatever complementary information the non-circuit weights carry, so stripping it improves performance; on ImageNet-1K, with 1,000 classes and millions of training images, the complementary information becomes useful and the scaffolding cost is amortised over many more gradient steps. The primary carrier of transfer is fixed across regimes; the fraction of the total benefit it accounts for is regime-dependent.

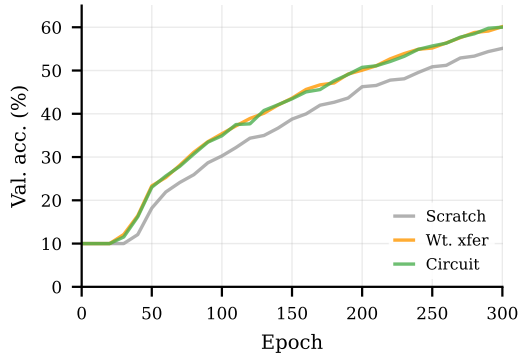


Figure 2: **Training dynamics** on Tiny-ImageNet. Circuit transfer (green) maintains a consistent advantage over scratch (grey) throughout 300 epochs.

Table 5: **Scale validation on ImageNet-1K** (ViT-Base, 300 epochs, DeiT recipe). Full warmup remains best in absolute accuracy; circuit transfer recovers 47% of the warmup gain (+0.81%), while plain attention-weight transfer recovers only 26%, isolating the contribution of SVD refactorisation over raw weight copying.

Method	Top-1 (%)	Δ Scratch	% Warmup Gain
Scratch	77.72	—	0
Full warmup	79.50	+1.78	100
Weight transfer (attn, L8–11)	78.17	+0.45	26
Circuit transfer (QK+OV, L8–11)	78.53	+0.81	47

Table 6: **Rank-truncated circuit transfer** (Tiny-ImageNet, ViT-Tiny, L8–11). “% Recov.” is normalised between $k=4$ (0%) and full rank (100%). Even half-rank preserves under half the benefit.

Rank k	Top-1	% Recov.
4	55.0	0
8	56.4	22
16	57.1	33
32	57.8	47
64 (full)	60.87	100

Table 7: **Decomposing the circuit** (Tiny-ImageNet, ViT-Tiny, L8–11). The OV circuit alone recovers nearly four times the QK circuit, and the combined effect exceeds the sum of parts.

Circuit	Top-1	Δ	% Gain
None (scratch)	56.45	—	0
QK only	57.15	+0.70	17
OV only	59.44	+2.99	71
QK + OV	60.87	+4.42	105

4 Mechanism and Universality

Section 3 established that attention circuits in late-layer heads are the primary carrier of pretraining transfer. We now turn from *what* transfers to *how* circuits arise during pretraining and *whether* the same structures appear in standard self-supervised methods.

4.1 The mechanism: OV dominance and two-phase emergence

The QK and OV products serve distinct roles—“where to attend” versus “what to extract”—and need not contribute equally to transfer. We isolate each by transferring it alone, leaving the other at Xavier initialisation, and find a striking asymmetry (Table 7): the OV circuit alone recovers 71% of the full pretraining gain while the QK circuit alone recovers only 17%, a four-fold dominance. Combined, the two circuits recover 105%, exceeding the additive prediction of 88% and indicating synergy: a properly initialised QK circuit becomes more effective when the OV circuit is also in place. This asymmetry runs against a strong default in the interpretability literature, which has predominantly focused on QK structure—positional biases, head-level specialisation [Clark et al., 2019, Voita et al., 2019]—because the QK circuit determines *where* a head looks and is more visually intuitive. Our results suggest that, for pretraining transfer at least, the more important question is *what* a head extracts. The asymmetry is also consistent with the rank-truncation behaviour of the circuits. Table 6 reports what happens when we truncate the SVD of M_h and T_h to a low rank before refactorising: even retaining the top-32 singular components (half of full rank) recovers only 47% of the full-rank performance, despite the top components carrying the majority of the Frobenius-norm energy. The transferable computation is therefore broad in information content—spread across many singular components, especially on the OV side—and is poorly approximated by low-rank surrogates such as those used in LoRA-style adaptation [Hu et al., 2022]. Notably, the rank-truncation curve is sub-linear: doubling the rank from $k=8$ to $k=16$ adds only 11 percentage points of gain, and recovering the full benefit requires all 64 components—the transferable computation exploits the entire rank of each product matrix.

A complementary spatial picture appears when we examine where in the network these properties accumulate. Figure 3 reports per-layer Q-K weight correlation, QK effective rank, and OV effective rank for the DYCK-pretrained ViT-Tiny. The Q-K correlation is small in early blocks and rises steadily towards block 10, peaking at 0.25; QK effective rank stays compressed across all layers (23–28, versus 58 at random initialisation) but is lowest in the late blocks; OV effective rank is

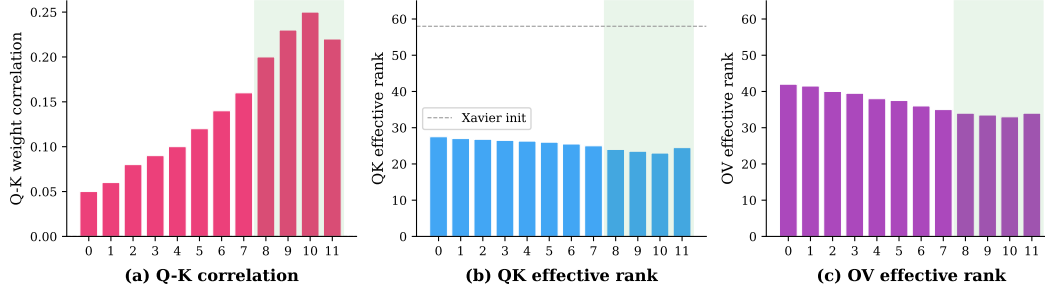


Figure 3: **Per-layer circuit metrics** for the DYCK-pretrained ViT-Tiny. The shaded region marks blocks 8–11, where circuit transfer is applied. All three metrics show that circuit structure concentrates in the late layers: Q-K correlation rises towards block 10 (a), QK effective rank is most compressed in the same range (b), and OV effective rank is consistently larger than QK rank, indicating a richer value-space structure (c).

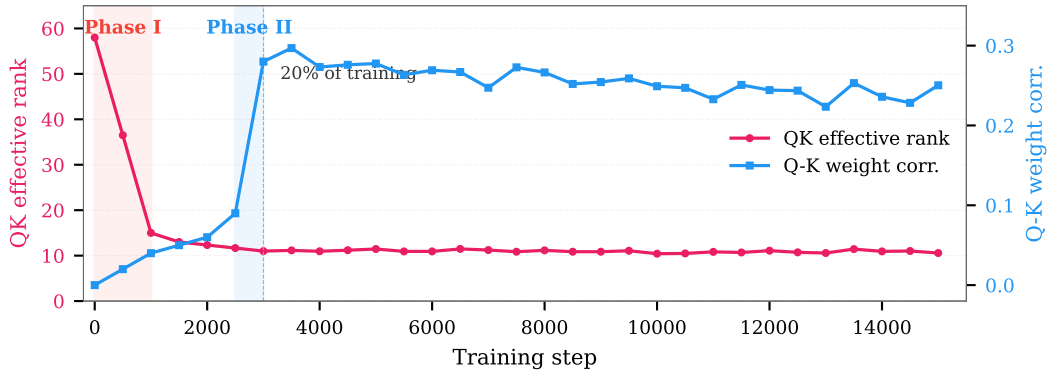


Figure 4: **Circuit emergence has two phases.** Phase I (steps 0–1,000): the QK effective rank collapses from 58 to 11. Phase II (steps 2,500–3,000): the Q-K weight correlation jumps from 0.09 to 0.28 in a single 500-step interval. Both metrics stabilise after step 3,000 (20% of total training).

consistently higher than QK, indicating a richer value-space structure. The shaded region (blocks 8–11) marks the layer range used by circuit transfer, which coincides with the strongest circuit signature on every metric—giving an independent, model-internal justification for the late-layer locus identified empirically by the elimination study of Section 3.3.

If circuits are the locus of transfer, when do they form? We track circuit-level metrics across 31 checkpoints saved every 500 steps during DYCK pretraining (steps 0–15,000), focusing on the QK effective rank [Roy and Vetterli, 2007] and the Pearson correlation between flattened W_Q and W_K matrices in blocks 8–11. Two distinct phases appear in Figure 4. Phase I, spanning the first 1,000 steps, is a rapid collapse of the QK effective rank from 58 (full rank at Xavier initialisation) to 11: the model learns to *focus*, compressing the attention computation onto roughly an eighth of the available subspace. Phase II, between steps 2,500 and 3,000, is a phase transition in the Q-K weight correlation: from 0.09 to 0.28 within a single 500-step interval. W_Q and W_K , which begin independent under Xavier initialisation, develop strong statistical coupling—the model learns to *align*. Both metrics stabilise after step 3,000, corresponding to roughly 20% of the total pretraining budget, and the remaining 80% refines task-specific behaviour without altering the structural properties on which transfer depends. These dynamics also explain why the spectral-blueprint controls of Section 3.3 fail: preserving each weight matrix’s singular values randomises the directions, which destroys exactly the W_Q - W_K alignment that emerges in Phase II.

4.2 Universality across pretraining methods

DYCK pretraining is a deliberately controlled probe; the conclusions matter only if circuits also describe what standard image-pretrained ViTs learn. We test this by computing the same circuit-level

Table 8: **Circuit signatures across pretraining methods** (ViT-Base, averaged over heads). All methods produce the same qualitative pattern as DYCK—elevated Q-K correlation, reduced QK effective rank, decreased head diversity—which is absent at random initialisation.

Model	Q-K correlation	QK effective rank	Q diversity (°)
Random init	0.00	62.6	75.5
MAE	0.34	45.8–53.6	66.9–69.7
DINO	0.31	38.8–55.7	63.7–69.6
DINOv2	0.38	44.6–60.6	65.2–68.1
DYCK (ViT-Tiny)	0.22	26.3–27.5	50.0–54.1

metrics on publicly available ViT-Base checkpoints from MAE [He et al., 2022], DINO [Caron et al., 2021], and DINOv2 [Oquab et al., 2024]. All three exhibit the same qualitative pattern as DYCK (Table 8): substantially elevated W_Q - W_K correlation, compressed QK effective rank, and reduced head diversity. Random initialisation shows none of these properties. Methods that differ in objective (masked reconstruction, self-distillation, contrastive learning) and in data scale nonetheless converge to the same circuit-level signature. The DYCK signature is sharper—the lower QK effective rank and higher correlation ratio reflect the simplicity of bracket matching—but it is the same kind of signature, not a different kind. Taken together with the emergence dynamics above, this places the circuit framework on a common footing with mainstream pretraining: what pretraining transmits, regardless of pretext task, is a specific computational shape that can be installed into attention by way of either weights or, more directly, circuits.

This convergence has an intuitive explanation. The attention mechanism imposes a shared architectural bottleneck: regardless of the pretext objective, every head must learn to route information through the same $W_Q^\top W_K$ and $W_{\text{proj}} W_V$ products. Pretraining objectives differ in *what* the network must predict, but they all require late-layer heads to develop selective attention (low QK rank) and aligned query–key subspaces (high Q-K correlation) in order to route task-relevant information effectively. The circuit signature is therefore not a property of any particular objective but a consequence of the transformer architecture itself—pretraining merely accelerates the formation of structure that the architecture is predisposed to develop.

5 Conclusion

Limitations. Circuit transfer matches or exceeds full pretraining on medium-complexity benchmarks but recovers only 47% of the warmup gain on ImageNet-1K with ViT-Base. The precise nature of the residual benefit carried by the discarded non-circuit parameters—and whether it can be captured without copying raw weights—remains an open question. All experiments use isotropic Vision Transformers (ViT-Tiny and ViT-Base); whether the empirical findings—particularly OV dominance and the late-layer locus—extend to hierarchical architectures such as Swin Transformers [Liu et al., 2021] or to dense prediction tasks (detection, segmentation) has not been verified.

Through systematic elimination across 20+ ablation conditions, we identified attention circuits in late-layer heads as the primary carrier of pretraining transfer in Vision Transformers, and introduced *circuit transfer*—SVD refactorisation of the QK and OV product matrices into fresh weights that preserve each head’s computation while discarding source-task scaffolding. Circuit transfer matches or exceeds full pretraining on Tiny-ImageNet (60.87% vs. 60.65%) and CIFAR-100 (75.52% vs. 75.03%), and recovers 47% of the warmup gain at ImageNet-1K scale with ViT-Base. Decomposing the circuit revealed a four-fold OV dominance over the QK side; trajectory analysis showed that circuit structure forms in two phases within the first 20% of pretraining; and analysis of MAE, DINO, and DINOv2 confirmed that all major self-supervised methods install the same circuit-level signatures. Taken together, these results reframe pretraining as the installation of computational primitives—specific input-output functions encoded in attention head products—rather than the accumulation of feature-tuned weight values. They suggest that the right question for understanding transfer is not *which weights to copy* but *which computations to preserve*.

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451 A Reproducibility Details

452 All experiments use a fixed random seed (42) and are fully deterministic given the same hardware
453 configuration. Code is anonymously available at this [link](#). Pre-trained MAE, DINO, and DINOv2
454 checkpoints are the publicly released ViT-Base weights from the respective authors. DYCK-64 data
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775 **subjects**

776 Question: Does the paper describe potential risks incurred by study participants, whether
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778 approvals (or an equivalent approval/review based on the requirements of your country or
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